

5.1 Chemical Energetics (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.1 Chemical Energetics (A Level Only)
Difficulty	Medium

Time allowed: 60

Score: /49

Percentage: /100

Question 1a

Lattice energies are always negative showing that they represent exothermic changes.

i)

Explain what is meant by lattice energy.

[2]

ii)

Explain why lattice energy is an exothermic process.

[1]

Table 5.1

energy change	value / kJ mol^{-1}
standard enthalpy change of atomisation of potassium	+89
electron affinity of $\text{O}(\text{g})$	-141
electron affinity of $\text{O}^-(\text{g})$	+798
standard enthalpy change of formation of potassium oxide	-361
first ionisation energy of potassium	+418
second ionisation energy of potassium	+3070
first ionisation energy of oxygen	+1310
second ionisation energy of oxygen	+3390
$\text{O}=\text{O}$ bond energy (diatomic molecule)	+496
$\text{O}-\text{O}$ bond energy (polyatomic molecule)	+150

[3 marks]

Question 1b

i)
Use relevant data from Table 5.1 to calculate the lattice energy, $\Delta H_{\text{latt}}^{\circ}$, of potassium oxide, $\text{K}_2\text{O}(\text{s})$.
Show your working.

$$\Delta H_{\text{latt}}^{\circ} \text{ of } \text{K}_2\text{O}(\text{s}) = \dots\dots\dots \text{kJ mol}^{-1} [3]$$

ii)
State how $\Delta H_{\text{latt}}^{\circ} \text{Na}_2\text{O}(\text{s})$ differs from $\Delta H_{\text{latt}}^{\circ} \text{K}_2\text{O}(\text{s})$.
Indicate this by placing **one** tick (✓) in the appropriate box in Table 5.2.

Table 5.2

$\Delta H_{\text{latt}}^{\circ} \text{Na}_2\text{O}(\text{s})$ is less negative than $\Delta H_{\text{latt}}^{\circ} \text{K}_2\text{O}(\text{s})$	$\Delta H_{\text{latt}}^{\circ} \text{Na}_2\text{O}(\text{s})$ is the same as $\Delta H_{\text{latt}}^{\circ} \text{K}_2\text{O}(\text{s})$	$\Delta H_{\text{latt}}^{\circ} \text{Na}_2\text{O}(\text{s})$ is more negative than $\Delta H_{\text{latt}}^{\circ} \text{K}_2\text{O}(\text{s})$

Explain your answer.

[1]

[4 marks]

Question 2a

One use of pure crystals of lithium fluoride are in X-ray monochromators.

The Born–Haber cycle for lithium chloride is shown in Fig. 2.1

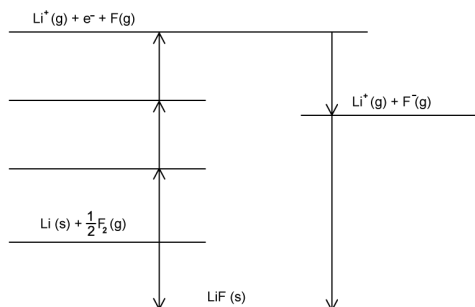


Fig. 2.1

- i)
Define the term enthalpy of atomisation.
- ii)
Explain why the enthalpy of atomisation of fluorine is positive.
- iii)
Complete the Born–Haber cycle for lithium fluoride in Fig 2.1 by adding the missing species on the lines.

[1]

[1]

[2]

[4 marks]

Question 2b

Use the data in the Table 2.1 and your completed Born–Haber cycle from part (a) to answer the questions below.

Table 2.1

Name of enthalpy change	Energy change / kJ mol^{-1}
$\text{Li(s)} \rightarrow \text{Li(g)}$	+216
$\text{Li(g)} \rightarrow \text{Li}^+(\text{g}) + \text{e}^-$	+520
$\text{F}_2(\text{g}) \rightarrow 2\text{F(g)}$	+158
$\text{F(g)} + \text{e}^- \rightarrow \text{F}^-(\text{g})$	-348
$\text{Li(s)} + \frac{1}{2}\text{F}_2(\text{g}) \rightarrow \text{LiF(s)}$	-594

i)

Calculate the enthalpy of lattice formation of lithium fluoride.

[2]

ii)

Explain and justify how the enthalpy of lattice formation of LiBr compares with that of LiF.

You must refer to the size of the ions in your answer.

[3]

[5 marks]

Question 2c

This question is about enthalpy changes in solution.

i)

Write the equation for the process showing the enthalpy of solution of potassium fluoride. Include state symbols in your answer.

[1]

ii)

Use the data in Table 2.2 to calculate the standard enthalpy of solution of potassium fluoride.

[2]

Table 2.2

Name of enthalpy change in solution	Enthalpy change (kJ mol^{-1})
Enthalpy of lattice dissociation potassium fluoride	+829
Enthalpy of hydration of potassium ions	-340
Enthalpy of hydration of fluoride ions	-504

[3 marks]

Question 2d

Explain why the value for the enthalpy of hydration, $\Delta H_{\text{hyd}}^{\ominus}$, of Group 1 ions increases from lithium to caesium.

[2 marks]

Question 3a

Calcium chloride, CaCl_2 , is an important industrial chemical used in refrigeration plants, for de-icing roads and for giving greater strength to concrete.

Use an equation to show what is meant by the lattice energy of calcium chloride.

[1 mark]

Question 3b

Explain how the lattice energies of the following salts compare in magnitude with that of calcium chloride.

- calcium fluoride
- calcium sulfide

[3 marks]

Question 3c

Use the data in Table 3.1 to calculate the lattice energy of CaCl_2 .

Table 3.1

standard enthalpy change of formation	-796 kJ mol^{-1}
standard enthalpy change of atomisation of Ca(s)	$+178 \text{ kJ mol}^{-1}$
electron affinity per mole of chlorine atoms	-349 kJ mol^{-1}
first ionisation energy of Ca	$+590 \text{ kJ mol}^{-1}$
second ionisation energy of Ca	$+1150 \text{ kJ mol}^{-1}$
standard enthalpy change of atomisation $\text{Cl}_2(\text{g})$	$+244 \text{ kJ mol}^{-1}$

[3 marks]

Question 3d

When a solution of CaCl_2 is added to a solution of the dicarboxylic acid, malonic acid, the salt calcium malonate is precipitated as a white solid.

The solid has the following composition by mass: Ca, 28.2%; C, 25.2%; H, 1.4%; O, 45.2%.

i)

Calculate the empirical formula of calcium malonate.

[2]

ii)

Suggest the structural formula of malonic acid.

[1]

[3 marks]

Question 4a

This question is about enthalpy change of hydration.

Explain what is meant by the term enthalpy change of hydration, $\Delta H^\theta_{\text{hyd}}$?

[1 mark]

Question 4b

i)

Write an equation that represents the $\Delta H^\theta_{\text{hyd}}$ of the Mg^{2+} ions.

[1]

ii)

Suggest a reason why the $\Delta H^\theta_{\text{hyd}}$ of the Mg^{2+} ions is greater than $\Delta H^\theta_{\text{hyd}}$ of the Ca^{2+} ion.

[1]

[2 marks]

Question 4c

The enthalpy change of solution for MgCl_2 , ($\text{MgCl}_2(\text{s})$), is represented by the following equation.



Describe the simple apparatus you could use, and the measurements you would make, in order to determine a value for ($\text{MgCl}_2(\text{s})$) in the laboratory.

[4 marks]**Question 4d**

Use the data in Table 4.1 to calculate the value for the enthalpy change of solution for magnesium chloride.

Table 4.1

$\Delta H^{\theta}_{latt} \text{MgCl}_2(\text{s})$	$-2526 \text{ kJ mol}^{-1}$
$\Delta H^{\theta}_{hyd}(\text{Mg}^{2+}(\text{g}))$	$-1890 \text{ kJ mol}^{-1}$
$\Delta H^{\theta}_{hyd}(\text{Cl}^{-}(\text{g}))$	-398 kJ mol^{-1}

[3 marks]

Question 5a

Magnesium reacts with fluorine to form magnesium fluoride.

Explain what is meant by the term first electron affinity.

[2 marks]

Question 5b

i)

Write an equation for the second electron affinity of fluorine.

[1]

ii)

Suggest why the second electron affinity for fluorine is endothermic.

[1]

[2 marks]

Question 5c

Determine the electron affinity of a fluorine atom, ΔH°_{EA} using Table 5.1.

[3]

Table 5.1

Name of enthalpy change	Energy change (kJ mol ⁻¹)
Enthalpy of atomisation of magnesium	+150
Second ionisation energy of magnesium	+1450
Enthalpy of formation of magnesium fluoride	-642
Lattice enthalpy of formation of magnesium fluoride	-2493

[1 mark]

Question 5d

Explain the trend in the first electron affinities going from fluorine to iodine.

[3 marks]



Model Answers: Medium

1a

a)

i) Lattice energy means:

- (The energy change) when **1 mole** of an ionic compound is formed; [1 mark]
- From **gaseous ions** (under standard conditions); [1 mark]

ii) Lattice energy is an exothermic process because:

- **Forming** an ionic bond; [1 mark]

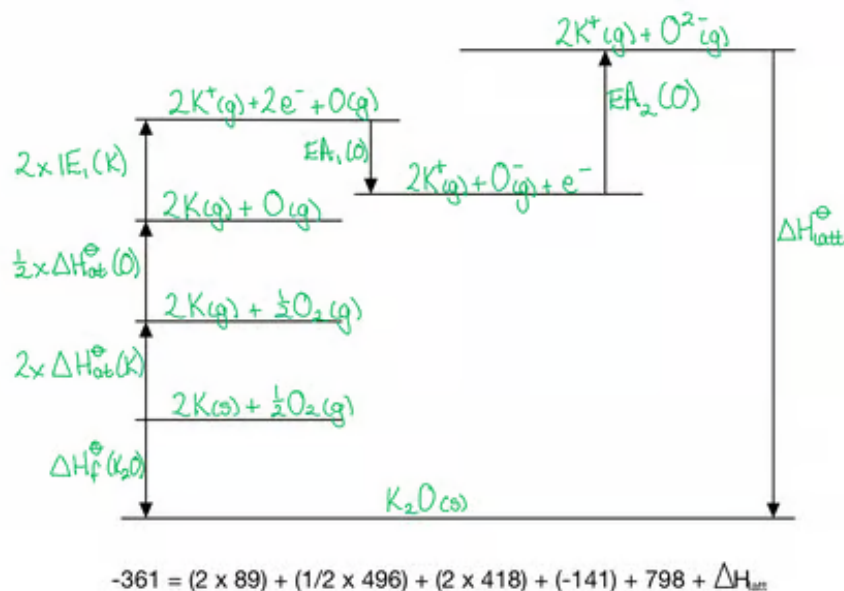
[Total: 3 marks]

- Lattice energy is one of the definitions of the different types of enthalpy changes that you need to learn
- It is worth investing time to know these definitions as they are easy marks to pick up and also will help you identify enthalpy changes from equations
- Lattice energy is always a large negative value as forming ionic bonds from gaseous ions releases a large amount of energy

1b

b)

i) The lattice energy, $\Delta H_{latt}^{\ominus}$, of potassium oxide, $K_2O(s)$ is:



- Extraction of relevant data $-361, -141, 89, 418, 496, 798$; [1 mark]
- Correct multipliers $2 \times (89); 2 \times (418); 0.5 \times (496)$; [1 mark]
- Correct sum and answer: $\text{LE} = -361 - 2(89) - 2(418) - 496 / 2 - (-141 + 798) = -2280$; [1 mark]

ii) The lattice energy of Na_2O differs from K_2O :

- Tick only in box 3 (more negative)
AND
 As sodium has a smaller ionic radius
AND
 So greater attraction between the ions; [1 mark]

[Total: 4 marks]

- When constructing the Born-Haber cycle for K_2O , remember that there are 2 moles of K so will require the enthalpy change of atomisation of K to be multiplied by 2
- There are $\frac{1}{2}$ mole of oxygen molecules so the atomisation of oxygen needs to be multiplied by $\frac{1}{2}$
- The 2 moles of potassium atoms will form $\text{K}^+ + \text{e}^-$ so the first ionisation energy of potassium will need to be multiplied by 2
- **Careful:** You are given the second ionisation energy of potassium but this is not required as it does not form a K^{2+} ion
- The most common errors in calculations of enthalpy changes are not multiplying the values given by the molar quantities shown in the equations and making

mistakes in rearranging the equation to find the lattice energy

- If you are not confident in rearranging equations, simplify the calculation as much as possible before you rearrange:
 - $-361 = (2 \times 89) + (\frac{1}{2} \times 496) + (2 \times 418) + (-141) + 798 + \Delta H_{latt}$
- Simplifying:
 - $-361 = 1919 + \Delta H_{latt}$
- Now rearrange - you may find it easier to leave ΔH_{latt} on the right-hand side and minus 1919 on the left-hand side:
 - $-361 - 1919 = \Delta H_{latt}$
 - $-2280 = \Delta H_{latt}$
- The magnitude in lattice energy is determined by both ionic radius and ionic charge
- When comparing lattice energies, always talk about the difference in ionic radius and/or ionic charge and how that affects the attraction between ions
- In this case, both cations have a 1+ charge, so it is the difference in ionic radius that will cause the difference
- As you go down the group, the ionic radius increases as there are more electron shells which means there is less attraction between the ion

2a

a)

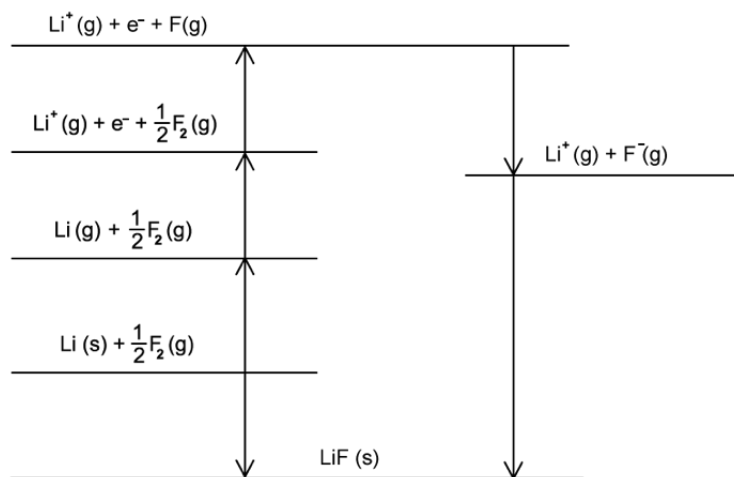
i) The enthalpy of atomisation is:

- The enthalpy change when one mole of gaseous atoms is formed from its elements under standard conditions; [1 mark]

ii) The enthalpy of atomisation of fluorine is positive because:

- Energy is always required to break any bonds between the atoms in the element; [1 mark]

iii) The completed Born-Haber cycle for lithium fluoride is:



- (Atomisation of Li): $\text{Li (g)} + \frac{1}{2}\text{F}_2\text{ (g)}$
Correct equation **AND** written on the correct line; [1 mark]
- (First ionisation of Li): $\text{Li}^+\text{ (g)} + \text{e}^- + \frac{1}{2}\text{F}_2\text{ (g)}$
Correct equation **AND** written on the correct line; [1 mark]

[Total: 4 marks]

- **Remember:** When drawing Born Haber cycles that the enthalpy of **atomisation** is forming **one mole of gaseous atoms** from an element in its standard state
 - So, for lithium, its state symbol has to change from (s) to (g)
 - For fluorine, it will atomise from $\frac{1}{2}\text{F}_2\text{ (g)}$ to F (g)
- You **always** have to include your state symbols and any electrons for each level in the cycle
- A good way to remember how to draw a Born Haber cycle is using the F.A.I.L technique:
 - F = Formation, A = Atomisation, I = Ionisation, L = Lattice Enthalpy
 - Start your cycles with the elements in their standard states with state symbols
 - Draw the lattice enthalpy of formation step going down to the ionic product
 - Work upwards in a clockwise direction for the remaining steps
 - Draw the atomisation steps, then the ionisation energy steps, then your electron affinity steps and lastly, lattice enthalpy of formation
- It doesn't matter if you write your atomisation step of the metal or nonmetal first, as long as they happen before the ionisation step

2b

b)

i) The lattice enthalpy of formation of lithium fluoride is:

- $\Delta H^\theta_{latt} \text{LiF} = -216 - 520 - \left(\frac{158}{2}\right) + 348 - 594$; [1 mark]
- $\Delta H^\theta_{latt} \text{LiF} = -1061 (\text{kJ mol}^{-1})$; [1 mark]

ii) The comparison between the lattice enthalpy of formation for NaBr and NaF is below:

- LiBr enthalpy of lattice formation will be less negative
OR
LiBr enthalpy of lattice formation will be less exothermic; [1 mark]
- Br^- ion is bigger (than F^-)
OR
 Br^- ion has more electron shells; [1 mark]
- So attraction of bromide ion (Br^-) to sodium ion (Na^+) is weaker; [1 mark]

[Total: 5 marks]

- The first thing to do when asked to calculate any value from a Born Haber Cycle is to insert the values on to the correct step on your cycle
 - That way you can clearly see the values associated and whether it is an exothermic step (arrow points downwards)
- The easiest way to work out which should be added and which should be subtracted is to draw a line around the outside of the cycle showing the alternative route from the lattice enthalpy of formation step
 - Any enthalpy changes which go in the same direction as the arrow should be added and any that go in the opposite direction should be subtracted
- Remember to also divide the atomisation of fluorine by two, as we only need one mole of fluorine atoms in LiF
 - The equation for atomisation of fluorine is $\frac{1}{2} \text{F}_2 (\text{g}) \rightarrow \text{F} (\text{g})$
- Also remember size of ions matter:
 - The larger the ion, the greater the distance between the centre of the oppositely charged ions, meaning the attraction between the ions will be weaker therefore there will be a smaller exothermic value for enthalpy of lattice formation

- LiBr enthalpy of lattice formation will be less negative

OR

LiBr enthalpy of lattice formation will be less exothermic; [1 mark]

- Br⁻ ion is bigger (than F⁻)

OR

Br⁻ ion has more electron shells; [1 mark]

- So attraction of bromide ion (Br⁻) to sodium ion (Na⁺) is weaker; [1 mark]

[Total: 5 marks]

- The first thing to do when asked to calculate any value from a Born Haber Cycle is to insert the values on to the correct step on your cycle
 - That way you can clearly see the values associated and whether it is an exothermic step (arrow points downwards)
- The easiest way to work out which should be added and which should be subtracted is to draw a line around the outside of the cycle showing the alternative route from the lattice enthalpy of formation step
 - Any enthalpy changes which go in the same direction as the arrow should be added and any that go in the opposite direction should be subtracted
- Remember to also divide the atomisation of fluorine by two, as we only need one mole of fluorine atoms in LiF
 - The equation for atomisation of fluorine is $\frac{1}{2} \text{F}_2 (\text{g}) \rightarrow \text{F} (\text{g})$
- Also remember size of ions matter:
 - The larger the ion, the greater the distance between the centre of the oppositely charged ions, meaning the attraction between the ions will be weaker therefore there will be a smaller exothermic value for enthalpy of lattice formation

2c

c)

i) The equation showing the enthalpy of solution for potassium fluoride is;

- $\text{KF} (\text{s}) \rightarrow \text{K}^+ (\text{aq}) + \text{F}^- (\text{aq})$; [1 mark]

ii) The calculation for enthalpy of solution for silver fluoride is;

$$\Delta H_{\text{sol}}^{\theta} \text{KF} = \Delta H_{\text{latt}}^{\theta} \text{KF} + (\Delta H_{\text{hyd}}^{\theta} \text{K}^+) + (\Delta H_{\text{hyd}}^{\theta} \text{F}^-)$$

- $\Delta H_{sol}^\theta \text{KF} = 829 - 340 - 504$; [1 mark]
- $\Delta H_{sol}^\theta \text{KF} = -15$ (kJ mol⁻¹); [1 mark]

[Total: 3 marks]

- When writing equations to represent enthalpy of solution it is **essential** that you include **state symbols** as well as including the charges on the ions
 - Even if the exam question doesn't ask, you **must** include state symbols throughout this topic to achieve the marks
- You could include H₂O over the arrow or write the equation as an equilibrium and you will still get the marks
- When calculating enthalpy of solution remember;
 - Breaking up the lattice (dissociation) is an endothermic process
 - Creating the lattice (formation) is an exothermic process
 - The values are exactly the same in terms of energy transfers, but the signs are different depending on if it is lattice dissociation or formation

2d

d) The enthalpy of hydration, ΔH_{hyd}^θ , of Group 1 ions increases from lithium to caesium because:

- Ionic radius increases down the group; [1 mark]
- Therefore, attraction between the larger ion and δ^- O atoms in the water / H₂O molecule decreases; [1 mark]

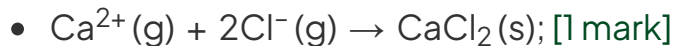
[Total: 2 marks]

- **Careful:** The enthalpy of hydration for the Group 1 ions becomes more positive or more endothermic down the group
- Therefore the value will increase as shown by the following ΔH_{hyd}^θ values
 - Li⁺ = -538 kJ mol⁻¹
 - Na⁺ = -424 kJ mol⁻¹
 - K⁺ = -340 kJ mol⁻¹
 - Rb⁺ = -315 kJ mol⁻¹
 - Cs⁺ = -291 kJ mol⁻¹
- A smaller ion gives a more exothermic hydration enthalpy as the attraction between the ion and polar water molecules is stronger
- A larger ion gives a less exothermic hydration enthalpy as the attraction between

the ion and polar water molecules is weaker

3a

a) An equation to show what is meant by the lattice energy of calcium chloride is:



[Total: 1 mark]

- Lattice energy is defined as the enthalpy change when 1 mole of ionic compound is formed from its **gaseous** ions under standard conditions
- Your equation should therefore show one mole of calcium chloride being formed
- Don't forget to balance the chloride ions
- State symbols **must** be correct to score the mark

3b

b) Explain how the lattice energies of the following salts compare in magnitude with that of calcium chloride:

- Both have larger lattice enthalpies (than CaCl_2)

OR

Lattice energies are more exothermic; [1 mark]

- F^{-} is smaller than Cl^{-} ; [1 mark]
- S^{2-} is more highly charged than Cl^{-} ; [1 mark]

[Total: 3 marks]

- **Remember:** Two factors affect the lattice energy
 - Charge of the ions- As the charge of the ions increases, the lattice energy gets more exothermic because of stronger electrostatic attraction between the oppositely charged ions
 - Radius of the ions- As the radius of the ion increases, the lattice energy gets less exothermic because there is more distance between the centre of the ions reducing the attraction

3c

c) The lattice energy of CaCl_2 is:

- Lattice energy = $-796 - (178 + 590 + 1150 + 244 + (2 \times 349))$; [1 mark]
- -2260 ; [1 mark]
- Negative answer; [1 mark]

[Total: 3 marks]

- To calculate the lattice energy use:
 - $\Delta H_{latt}^\theta = \Delta H_f^\theta - \Delta H_l^\theta$
 - $\Delta H_l^\theta = \Delta H_{at}^\theta + \Delta H_{at}^\theta + IE + EA$
- Lattice enthalpies are always exothermic so the value should be negative

3d

d)

i) The empirical formula of calcium malonate is:

- | Ca | C | H | O | |
|-----------------------------|---------------------------|-------------------------|----------------------------|------------|
| $\frac{28.2}{40.1} = 0.703$ | $\frac{25.2}{12.0} = 2.1$ | $\frac{1.4}{1.0} = 1.4$ | $\frac{45.2}{16.0} = 2.83$ | ; [1 mark] |
- $\text{CaC}_3\text{H}_2\text{O}_4$; [1 mark]

ii) The structural formula of malonic acid is:

- $\text{CH}_2(\text{CO}_2\text{H})_2$; [1 mark]

[Total: 3 marks]

- To find the empirical formula divide the percentage of each element by its relative atomic mass
- Then find the ratio by dividing each answer by the smallest value, in this case 0.703
- For part (ii)
 - $\text{CaC}_3\text{H}_2\text{O}_4$ is the empirical formula of the salt formed between malonic acid and calcium chloride
 - It consists of Ca^{2+} (from the CaCl_2) and $\text{C}_3\text{H}_2\text{O}_4^{2-}$ (from the malonic acid)
 - You are told that malonic acid is a dicarboxylic acid so this accounts for 2 x COO from the $\text{C}_3\text{H}_2\text{O}_4^{2-}$

- This leaves a $-\text{CH}_2$ group so the ion from the malonic acid must be $\text{CH}_2(\text{COO})_2^{2-}$
- When the malonic acid reacts with the CaCl_2 it loses the H atoms from the COOH groups
- The structure of malonic acid is therefore $\text{CH}_2(\text{COOH})_2$

4a

a) The term enthalpy change of hydration, $\Delta H^\theta_{\text{hyd}}$ means:

- The enthalpy change when 1 mole of gaseous ions dissolves in water; [1 mark]

[Total: 1 mark]

- This is one of the many enthalpy definitions you must learn
- It is important to state the ions are in their gaseous form or you will not score the mark

4b

b)

i) An equation that represents the $\Delta H^\theta_{\text{hyd}}$ of the Mg^{2+} ions is:

- $\text{Mg}^{2+}(\text{g}) \rightarrow \text{Mg}^{2+}(\text{aq})$; [1 mark]

ii) The $\Delta H^\theta_{\text{hyd}}$ of the Mg^{2+} ions is greater than $\Delta H^\theta_{\text{hyd}}$ of the Ca^{2+} ion because:

- Mg^{2+} has a smaller radius / size compared to Ca^{2+}
OR
 Mg^{2+} has greater charge density than Ca^{2+} ; [1 mark]

[Total: 2 marks]

- You would also score the mark for part (i) for writing: $\text{Mg}^{2+} + 6\text{H}_2\text{O} \rightarrow [\text{Mg}(\text{H}_2\text{O})_6]^{2+}$
- When an ionic substance dissolves in water, bonds form between the water molecules and the ions
 - The δ^- oxygen atoms in water molecules are attracted to the positive ions
 - The δ^+ hydrogen atoms in water molecules are attracted to the negative ions

- This attraction is stronger for ions with smaller ionic radii so the enthalpy of hydration is more exothermic

4c

c) The simple apparatus you could use, and the measurements you would make, in order to determine a value for $(\text{MgCl}_2(\text{s}))$ in the laboratory:

- Measure (know volume / mass of) water into a calorimeter; [1 mark]
- Take the temperature of the water; [1 mark]
- Weight out known mass of MgCl_2 ; [1 mark]
- Take final / highest temperature

OR

Record temperature change; [1 mark]

[Total: 4 marks]

- Calorimetry can be used to measure enthalpy change of solution
- Once the above experiment has been conducted you would calculate the energy change (q) using $q = mc\Delta T$
- The energy change and mass of magnesium chloride can then be used to calculate the enthalpy change per mole of magnesium chloride
- Make sure your method specifies that you will measure a 'known' volume / mass of water and magnesium chloride as these are the control variables which will help you obtain valid results

4d

d) The value for the enthalpy change of solution for magnesium chloride is:

- $\Delta H^\theta_{\text{sol}} = \Delta H^\theta_{\text{hyd}}(\text{Mg}^{2+}) + 2 \times \Delta H^\theta_{\text{hyd}}(\text{Cl}^-) - \Delta H^\theta_{\text{latt}}$; [1 mark]
- $\Delta H^\theta_{\text{sol}} = (-1890) + 2 \times (-398) - (-2526)$; [1 mark]
- $\Delta H^\theta_{\text{sol}} = -160 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 3 marks]

- You should know the equation: $\Delta H^\theta_{\text{latt}} + \Delta H^\theta_{\text{sol}} = \Delta H^\theta_{\text{hyd}}$
- Rearrange this to find $\Delta H^\theta_{\text{sol}}$
 - $\Delta H^\theta_{\text{sol}} = \Delta H^\theta_{\text{hyd}} - \Delta H^\theta_{\text{latt}}$
- You must make sure you multiple the $\Delta H^\theta_{\text{hyd}}$ for the chloride ions by two as two

ions are formed in solution

5a

a) The term first electron affinity is:

- The enthalpy change when 1 mole of electrons is added; [1 mark]
- To 1 mole of gaseous atoms to form 1 mole of gaseous ions with a single negative charge under standard conditions; [1 mark]

[Total: 2 marks]

- It is important to get the exact wording or you will not score the marks
- Make sure you learn all the enthalpy definitions

5b

b)

i) An equation for the second electron affinity of fluorine is:

- $F^{-}(g) + e^{-} \rightarrow F^{2-}(g)$; [1 mark]

ii) The second electron affinity for fluorine is endothermic because:

- Energy is needed to add 1 mole of electrons to (1 mole of) positive ions; [1 mark]

[Total: 2 marks]

- The second electron affinity is the enthalpy change when 1 mole of electrons is added to 1 mole of gaseous 1- ions to form 1 mole of gaseous 2- ions under standard conditions
- Your equation for second electron affinity should therefore start with 1 mole of fluoride ions
- You will usually be asked to write the equation for second electron affinity as part of a Born-Haber cycle for elements that form compounds consisting of negative ions with a 2- charge, for example oxide ions
- You **must** include state symbols in your equation even if the question doesn't explicitly tell you to or you will not score the marks
- **Remember:** The first electron affinity is always exothermic, so negative with energy released when 1 mole of electrons are added but the second and third

are endothermic (or positive) because energy is needed to force 1 mole of electrons onto the already negative ions

5c

c) The electron affinity value for fluorine is:

$$\Delta H_f^\theta \text{MgF}_2 = \Delta H_{at}^\theta \text{Mg} + \Delta H_{IE1}^\theta \text{Mg} + \Delta H_{IE2}^\theta \text{Mg} + \Delta H_{BE}^\theta \text{F} + [2 \times \Delta H_{EA1}^\theta \text{F}] + \Delta H_{latt}^\theta \text{MgF}_2$$

OR

$$-\Delta H_f^\theta \text{MgF}_2 + \Delta H_{at}^\theta \text{Mg} + \Delta H_{IE1}^\theta \text{Mg} + \Delta H_{IE2}^\theta \text{Mg} + \Delta H_{BE}^\theta \text{F} = -[2 \times \Delta H_{EA1}^\theta \text{F}] - \Delta H_{latt}^\theta \text{MgF}_2$$

OR

$$2 \times \Delta H_{EA1}^\theta \text{F} = \Delta H_f^\theta \text{MgF}_2 - \Delta H_{at}^\theta \text{Mg} - \Delta H_{IE1}^\theta \text{Mg} - \Delta H_{IE2}^\theta \text{Mg} - \Delta H_{BE}^\theta \text{F} - \Delta H_{latt}^\theta \text{MgF}_2; [1 \text{ mark}]$$

$$2 \times \Delta H_{EA1}^\theta \text{F} = -642 - 150 - 738 - 1450 - 159 - (-2493) = -646; [1 \text{ mark}]$$

$$\Delta H_{EA1}^\theta \text{F} = \frac{-646}{2} = -323 \text{ (kJ mol}^{-1}\text{)}; [1 \text{ mark}]$$

[Total: 5 marks]

- You can sketch a simple cycle in the exam and draw your 'direct route' and 'indirect route' to help with your calculations
 - Any enthalpy changes which go in the **same direction** as the arrow you **add** and any that go in the **opposite direction** you **subtract**
- Careful:** you need to multiply the atomisation of F by 2 to allow for 2 moles of fluorine atoms in MgF_2

5d

d) The trend in the first electron affinities going from fluorine to iodine is:

- (Electron affinity) decreases; [1 mark]
- There is more shielding by inner electrons

AND

The number of shells increases / ionic radius increases; [1 mark]

- There is less attraction between the electron being gained and the positively charged nucleus; [1 mark]

[Total: 3 marks]